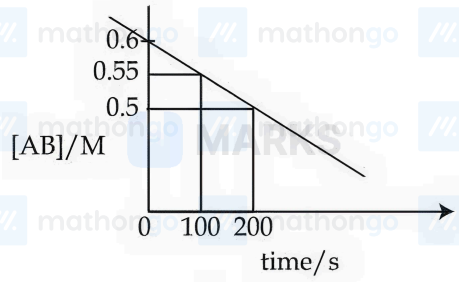
Q1. JEE Main 2026 (21 January Shift 2)

Decomposition of A is a first order reaction at $T(K)$ and is given by $A(g) \longrightarrow B(g) + C(g)$. In a closed 1 L vessel, 1 bar $A(g)$ is allowed to decompose at $T(K)$. After 100 minutes, the total pressure was 1.5 bar. What is the rate constant $(\text{in} \min^{-1})$ of the reaction ? $(\log 2 = 0.3)$

(1) 6.9×10^{-3}
(2) 6.9×10^{-4}
(3) 6.9×10^{-1}
(4) 6.9×10^{-2}

Q2. JEE Main 2026 (23 January Shift 1)

For the thermal decomposition of reactant $AB(g)$, the following plot is constructed.



The half life of the reaction is ' x' min. $x =$ ____ min. (Nearest integer)

Q3. JEE Main 2026 (28 January Shift 2)

$A \longrightarrow B$ (first reaction)

$C \longrightarrow D$ (second reaction)

Consider the above two first-order reactions. The rate constant for first reaction at 500 K is double of the same at 300 K. At 500 K, 50% of the reaction becomes complete in 2 hour. The activation energy of the second reaction is half of that of first reaction. If the rate constant at 500 K of the second reaction becomes double of the rate constant of first reaction at the same temperature; then rate constant for the second reaction at 300 K is ____ $\times 10^{-1}$ hour $^{-1}$ (nearest integer).

Q4. JEE Main 2026 (24 January Shift 2)

The half-life of ^{65}Zn is 245 days. After x days, 75% of original activity remained. The value of x in days is ____. (Nearest integer)

(Given: $\log 3 = 0.4771$ and $\log 2 = 0.3010$)

Q5. JEE Main 2026 (22 January Shift 2)

Correct statements regarding Arrhenius equation among the following are :

A. Factor $e^{-Ea/RT}$ corresponds to fraction of molecules having kinetic energy less than Ea.

B. At a given temperature, lower the Ea, faster is the reaction.

C. Increase in temperature by about 10°C doubles the rate of reaction.

D. Plot of $\log k$ vs $\frac{1}{T}$ gives a straight line with slope $= -\frac{Ea}{R}$.

Choose the correct answer from the options given below :

(1) B and D Only
(2) A and B Only
(3) B and C Only
(4) A and C Only

Q6. JEE Main 2026 (22 January Shift 2)

Consider $A \xrightarrow{k_1} B$ and $C \xrightarrow{k_2} D$ are two reactions. If the rate constant (k_1) of the $A \longrightarrow B$ reaction can be expressed by the following equation $\log_{10} k = 14.34 - \frac{1.5\times10^4}{T/K}$ and activation energy of $C \longrightarrow D$ reaction (Ea_2) is $\frac{1}{5}$ th of the $A \longrightarrow B$ reaction (Ea_1), then the value of (Ea_2) is ____ $kJmol^{-1}$. (Nearest Integer)

Q7. JEE Main 2026 (22 January Shift 1)

$A \rightarrow$ product (First order reaction). Three sets of experiment were performed for a reaction under similar experimental conditions:

Run 1 $\Rightarrow$ 100 mL of 10 M solution of reactant A

Run 2 $\Rightarrow$ 200 mL of 10 M solution of reactant A

Run 3 $\Rightarrow$ 100 mL of 10 M solution of reactant A + 100 mL of H_2O added.

The correct variation of rate of reaction is

(1) Run 1 < Run 2 < Run 3

(2) Run 3 < Run 1 < Run 2

(3) Run 3 < Run 1 = Run 2

(4) Run 1 = Run 2 = Run 3

Q8. JEE Main 2026 (22 January Shift 1)

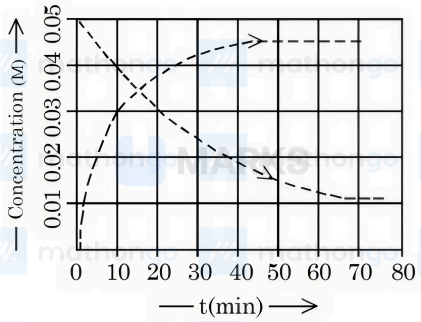
The temperature at which the rate constants of the given below two gaseous reactions become equal is ____ K. (Nearest integer)

$X \longrightarrow Y \quad k_1 = 10^6 e^{\frac{-30000}{T}}$

$P \longrightarrow Q \quad k_2 = 10^4 e^{\frac{-24000}{T}}$

Given: $\ln 10 = 2.303$

Q9. JEE Main 2026 (23 January Shift 2)



Given above is the concentration vs time plot for a dissociation reaction : $A \rightarrow nB$. Based on the data of the initial phase of the reaction (initial 10 min), the value of n is ____.

(1) 3 (2) 2 (3) 4 (4) 5

Q10. JEE Main 2026 (23 January Shift 2)

Observe the following reactions at $T(K)$.

I. $A \rightarrow$ products.

II. $5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$

Both the reactions are started at 10.00 am. The rates of these reactions at 10.10 am are same. The value of $-\frac{\Delta[Br^-]}{\Delta t}$ at 10.10 am is 2×10^{-4} mol L^{-1} min^{-1}. The concentration of A at 10.10 am is 10^{-2} mol L^{-1}. What is the first order rate constant (in min^{-1}) of reaction I ?

(1) 10^{-2} (2) 10^{-3} (3) 2×10^{-3} (4) 4×10^{-3}

Q11. JEE Main 2026 (21 January Shift 1)

Pre-exponential factors of two different reactions of same order are identical. Let activation energy of first reaction exceeds the activation energy of second reaction by 20 kJ mol^{-1}. If k_1 and k_2 are the rate constants of first and second reaction respectively at 300 K, then $\ln \frac{k_2}{k_1}$ will be ____. (nearest integer) $[R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}]$

Q12. JEE Main 2026 (28 January Shift 1)

An organic compound undergoes first order decomposition. The time taken for decomposition to $\left(\frac{1}{8}\right)^{th}$ and $\left(\frac{1}{10}\right)^{th}$ of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively. What is the value of $\frac{t_{1/8}}{t_{1/10}} \times 10$? $(\log 2 = 0.3)$

(1) 3 (2) 9 (3) 0.9 (4) 30

Q13. JEE Main 2026 (24 January Shift 1)

At 27°C in presence of a catalyst, activation energy of a reaction is lowered by 10 kJ mol^{-1}. The logarithm of ratio of $\frac{k \text{ (catalysed)}}{k \text{ (uncatalysed)}}$ is....

(Consider that the frequency factor for both the reactions is same)

(1) 1.741 (2) 17.41 (3) 3.482 (4) 0.1741

ANSWERS AND SOLUTIONS

1. (1)	**2.** 10	**3.** 5	**4.** 102	**5.** (3)	**6.** 57	**7.** (3)	**8.** 1303
9. (1)	**10.** (4)	**11.** 8	**12.** (2)	**13.** (1)			

1. **(1)** For the reaction $A(g) \rightarrow B(g) + C(g)$, initially pressure of A is 1 bar.

Initially: total pressure = 1 bar.

After 100 min: total pressure = 1.5 bar.

Using initial and final pressures, if x bar of A decomposes, then A becomes (1-x), B = x, C = x.

Total pressure = 1-x+x+x = 1+x = 1.5, so x = 0.5 bar.

For a first-order reaction: $k = \frac{2.303}{t}\log\frac{[A]_0}{[A]}$.

Using pressures instead: $k = \frac{2.303}{100}\log\frac{1}{0.5}$

$= \frac{2.303}{100} \times \log 2 = \frac{2.303\times0.3}{100} = \frac{0.691}{100}$

$= 6.91 \times 10^{-3}\ \text{min}^{-1} \approx 6.9 \times 10^{-3}\ \text{min}^{-1}$

2. **(10)** From the concentration vs. time graph, the thermal decomposition of AB(g) shows a linear decrease in concentration, indicating zeroth-order kinetics.

Initial concentration: $[AB]_0$ = 0.6 M at t = 0 s.

Reading from the graph: [AB] = 0.5 M at t = 200 s.

Rate of decomposition: Δ[AB]/Δt = (0.6 - 0.5) M / 200 s = 0.0005 M/s.

For zeroth-order kinetics: $[AB] = [AB]_0 - kt$.

Half-life occurs when $[AB] = [AB]_0/2$ = 0.3 M.

$0.3 = 0.6 - 0.0005 \times t_{1/2}$.

$t_{1/2}$ = 0.3 / 0.0005 = 600 s = 10 min.

3. **(5)** For $A \xrightarrow{K_1} B$

$$\ln(2) = \frac{E_{a_1}}{R}\left[\frac{1}{300} - \frac{1}{500}\right]$$

$$E_{a_1} = \frac{\ln 2 \times R \times 1500}{2}$$

$$E_{a_2} = \frac{E_{a_1}}{2} = \frac{\ln 2 \times R \times 1500}{4}$$

$$(K_1)_{\text{at 500 K}} = \frac{\ln 2}{2}$$

$$(K_2)_{\text{at 500 K}} = \ln 2$$

Now for $C \xrightarrow{K_2} D$

$$\ln\left[\frac{(K_2)_{\text{at 500K}}}{(K_2)_{\text{at 300K}}}\right] = \left(\frac{\ln 2 \times R \times 1500}{4}\right) \times \frac{1}{R} \times \left[\frac{1}{300} - \frac{1}{500}\right]$$

$$(K_2)_{\text{at 300 K}} = \frac{\ln 2}{\sqrt{2}} = 0.49$$

$(K_2)_{\text{at 300 K}} = 4.9 \times 10^{-1}$.
So, answer is 5.

4. **(102)** Using radioactive decay: $A = A_0\left(\frac{1}{2}\right)^{t/t_{1/2}}$.
With 75% activity remaining: $0.75 = \left(\frac{1}{2}\right)^{x/245}$.
Taking log base 10: $\log(0.75) = \frac{x}{245}\log(0.5)$.
Calculate: $\log(0.75) = \log 3 - \log 4 = 0.4771 - 2(0.3010) = -0.1249$ and $\log(0.5) = -0.3010$.
Thus $-0.1249 = \frac{x}{245} \times (-0.3010)$, giving $x = \frac{0.1249 \times 245}{0.3010} = 101.66 \approx 102$ days (nearest integer).

5. **(3)** The Arrhenius equation is given by $k = Ae^{-E_a/RT}$.
Statement A: The factor $e^{-E_a/RT}$ represents the fraction of molecules having kinetic energy equal to or greater than the activation energy E_a. Thus, statement A is incorrect.
Statement B: From the equation, as E_a decreases, the value of k increases, leading to a faster reaction rate. Thus, statement B is correct.
Statement C: For many reactions, the temperature coefficient is approximately 2, meaning the rate of reaction doubles for every $10^\circ C$ rise in temperature. Thus, statement C is correct.
Statement D: Taking the natural logarithm of the Arrhenius equation, $\ln k = \ln A - \frac{E_a}{RT}$. Converting to base 10, $\log k = \log A - \frac{E_a}{2.303RT}$. A plot of $\log k$ vs $\frac{1}{T}$ gives a straight line with slope $= -\frac{E_a}{2.303R}$. The statement says slope $= -\frac{E_a}{R}$, which is only true for a $\ln k$ vs $\frac{1}{T}$ plot. Thus, statement D is incorrect.
Therefore, statements B and C are correct.

6. **(57)** From the Arrhenius equation:
$\log_{10} k = \log_{10} A - \frac{E_a}{2.303RT}$.
Comparing with given equation $\log_{10} k_1 = 14.34 - \frac{1.5 \times 10^4}{T}$, we get: $\frac{E_{a_1}}{2.303R} = 1.5 \times 10^4$ K

$E_{a_1} = 1.5 \times 10^4 \times 2.303 \times 8.314$
$= 1.5 \times 10^4 \times 0.01913$
$= 287$ kJ/mol

Since $E_{a_2} = \frac{1}{5}E_{a_1}$:

$E_{a_2} = \frac{287}{5} = 57.4 \approx 57$ kJ/mol

7. **(3)** For a first-order reaction, rate = $k[A]$, where rate is directly proportional to reactant concentration.
Run 1: 100 mL of 10 M solution
Concentration = 10 M
Rate: $r_1 = k \times 10$
Run 2: 200 mL of 10 M solution
Concentration remains 10 M (increasing volume doesn't change concentration)

Rate: $r_2 = k \times 10$

Therefore $r_1 = r_2$

Run 3: 100 mL of 10 M solution + 100 mL of H_2O

Total moles of A = 10 M × 0.1 L = 1 mol

Total volume = 0.2 L

New concentration = 1 mol / 0.2 L = 5 M

Rate: $r_3 = k \times 5$

Therefore $r_3 < r_1 = r_2$

8. **(1303)** To find the temperature at which $k_1 = k_2$,

set the rate constant equations equal: $10^6 e^{-30000/T} = 10^4 e^{-24000/T}$.

Dividing both sides by 10^4: $100 = e^{-24000/T+30000/T} = e^{6000/T}$.

Taking natural logarithm: $\ln(100) = \frac{6000}{T}$.

Since $\ln(100) = 2\ln(10) = 2 \times 2.303 = 4.606$: $4.606 = \frac{6000}{T}$.

Solving for T: $T = \frac{6000}{4.606} = 1302.5$ K.

Rounding to the nearest integer gives T = 1303 K.

9. **(1)** $A \to nB$

At $t = 0$: $[A] = 0.05$ M, $[B] = 0$

At $t = 10$ min: $[A] = 0.04$ M, $[B] = 0.01 \times n$

From the graph, $[B]$ at $t = 10$ min $= 0.03$ M

$0.01 \times n = 0.03$

$n = 3$

10. **(4)** For reaction II:

$5Br^- + BrO_3^- + 6H^+ \to 3Br_2 + 3H_2O$,

The overall reaction rate is:

$r = -\frac{1}{5}\frac{\Delta[Br^-]}{\Delta t} = -\frac{1}{3}\frac{\Delta[Br_2]}{\Delta t}$.

Given $\frac{\Delta[Br^-]}{\Delta t} = 2 \times 10^{-4}$ mol L^{-1} min^{-1}

The reaction rate is $r = \frac{2\times10^{-4}}{5} = 4 \times 10^{-5}$ mol L^{-1} min^{-1}.

Since the rates of both reactions are equal at 10.10 am and reaction I is first-order:

$r = k[A]$, so $4 \times 10^{-5} = k \times 10^{-2}$, giving $k = 4 \times 10^{-3}$ min^{-1}.

11. **(8)** Given: $A_1 = A_2$, $E_{a1} = E_{a2} + 20$ kJ/mol, T = 300 K, R = 8.3 J K^{-1} mol^{-1}

Using Arrhenius equation: $k = Ae^{-E_a/RT}$

$\frac{k_2}{k_1} = \frac{e^{-E_{a2}/RT}}{e^{-E_{a1}/RT}} = e^{(E_{a1}-E_{a2})/RT}$

$E_{a1} - E_{a2} = 20$ kJ/mol = 20000 J/mol

$\frac{k_2}{k_1} = e^{\frac{20000}{8.3\times300}} = e^{\frac{20000}{2490}}$

$\ln\frac{k_2}{k_1} = \frac{20000}{2490} = 8.03 \approx 8$

12. (2) For first-order decomposition with rate equation $[A] = [A]_0 e^{-kt}$, the time for concentration to reduce to fraction f is: $t = \frac{1}{k}\ln\frac{1}{f} = \frac{\ln(1/f)}{k}$

For reduction to 1/8: $t_{1/8} = \frac{\ln 8}{k} = \frac{3\ln 2}{k}$

For reduction to 1/10: $t_{1/10} = \frac{\ln 10}{k}$

Therefore: $\frac{t_{1/8}}{t_{1/10}} = \frac{\ln 8}{\ln 10} = \frac{3\log 2}{\log 10} = \frac{3\times0.3}{1} = 0.9$

Thus: $\frac{t_{1/8}}{t_{1/10}} \times 10 = 0.9 \times 10 = 9$

13. (1) Using the Arrhenius equation: $k = Ae^{-E_a/RT}$

The ratio of rate constants is: $\frac{k_{catalysed}}{k_{uncatalysed}} = \frac{Ae^{-E_a^c/RT}}{Ae^{-E_a^u/RT}} = e^{(E_a^u - E_a^c)/RT} = e^{\Delta E_a/RT}$

Since the catalyst lowers activation energy by 10 kJ/mol: $\Delta E_a = 10$ kJ/mol = 10000 J/mol

At T = 27°C = 300 K, R = 8.314 J/(mol·K):

$\frac{k_c}{k_u} = e^{10000/(8.314\times300)} = e^{10000/2494.2} = e^{4.009}$

Taking logarithm base 10: $\log_{10}\frac{k_c}{k_u} = 4.009 \times \log_{10}(e) = 4.009 \times 0.4343 = 1.741$